

From flow fields to defensible labels

Learning outcomes: distinguish rotation from shear and compression; explain why shock and vortex labels may overlap; fit and judge a small learned detector without pixel leakage; separate weak-reference agreement from physical validation.

Suggested 90-minute class: 15 minutes of diagnostic controls, 20 minutes of physical framing, 30 minutes in the notebook, 15 minutes of evidence critique, and 10 minutes for the exit assessment. Prerequisites are velocity gradients, supervised learning, cylinder wakes and compressible-flow branches from Weeks 1, 2, 7 and 8.

Research basis: Ehsan Roohi, Physics-audited joint neural segmentation of shocks and vortex cores: cross-solver transfer and controlled airfoil--cylinder studies (author-supplied research manuscript, 2026). The source describes joint detection in airfoil and cylinder fields. Do not describe the manuscript as a published JCP article or invent a journal DOI.

The executable lab starts with an original CPU teaching analog. Its final section reads real airfoil and cylinder research fields through retained figures and a provenance ledger: six fresh forward passes with a frozen research checkpoint. No new CFD or research-model training is claimed. The analytic warm-up and research evidence remain explicitly separate.

Vorticity is necessary context, not a core label

Let A be the in-plane velocity-gradient tensor with rows (u_x, u_y) and (v_x, v_y) . Its symmetric part describes strain and its antisymmetric part describes local rotation. Planar vorticity is $\omega_z = v_x - u_y$; dilatation is $\theta = u_x + v_y$. A large derivative can also result from wall shear, a grid artifact or amplified sampling noise.

For the trace-free in-plane tensor, $Q_d = -(u_x - v_y)^2/4 - u_y v_x$. The imaginary part of its complex eigenvalues gives swirling strength: $\lambda_{ci} = \sqrt{\max(Q_d, 0)}$. In two dimensions these are algebraically linked, not two independent pieces of evidence. This is not the full three-dimensional vortex-classification problem.

Work three controls by hand: solid rotation $u=-y, v=x$ gives $\omega=2$ and $\lambda_{ci}=1$; simple shear $u=y, v=0$ gives $\omega=-1$ but $\lambda_{ci}=0$; isotropic compression $u=-x, v=-y$ gives $\theta=-2$ and no swirling strength. The notebook verifies these identities numerically.

These gradients are unchanged by constant velocity translations on a fixed grid. That does not make the detector objective under arbitrary time-dependent rotating frames.

A shock is more than a bright gradient

Density or pressure gradients indicate edges, not uniquely shocks. Contact layers, solid boundaries, expansion regions and numerical ringing can also produce strong signals. Compression, pressure rise and front geometry are complementary evidence. A velocity-gradient sensor alone cannot verify a gas-dynamic jump.

Across a locally planar stationary shock, audit the normal mass, momentum and total-energy fluxes: ρu_n ; $\rho u_n^2 + p$; and $\rho u_n(h + |u|^2/2)$. A moving front requires velocity relative to that front. Use upstream and downstream states outside the numerically thick transition; oblique fronts require a local normal and consistent orientation.

The teaching field uses a tanh compression layer superposed with rotation and shear. Its labels are known construction regions. The superposition is not a Navier-Stokes or Euler solution and is not required to satisfy these jump conditions. Call it a shock-like diagnostic control, not a verified physical shock.

Discussion: why would a sensor appear accurate if its own thresholded gradient map were also used as ground truth? Design a separate jump audit and a solid-wall negative control before proposing an improved classifier.

Overlapping structures need independent decisions

A vortex can intersect a shock. An exclusive softmax over background, shock and vortex prohibits this overlap by construction. Independent sigmoid foreground outputs allow two labels at one location; a background mask can then be derived from their complement. Ambiguous regions and exact solid geometry require explicit treatment rather than silent relabelling.

The research framework uses separate learned, physics-only and hybrid products, with task-restricted adaptation. If the shared encoder and the complete vortex pathway are frozen, changing only the shock branch preserves vortex outputs. Freezing the vortex decoder alone is insufficient when its input encoder continues to change. Output equality is a testable claim about computational dependencies, not a generic property of multitask learning.

For conflicting shared gradients g_s and g_v , projection methods can remove a component when their dot product is negative. Such methods can help one task and hurt another; retain both signed changes. Physical diagnostic channels do not by themselves constitute a PDE-residual loss or a PINN.

Our local MLP has eight primitive/derivative features and two independent output decisions. It teaches overlap and evaluation, but has no spatial encoder, dual convolutional decoder or research adaptation mechanism.

Freeze the experiment before opening the test

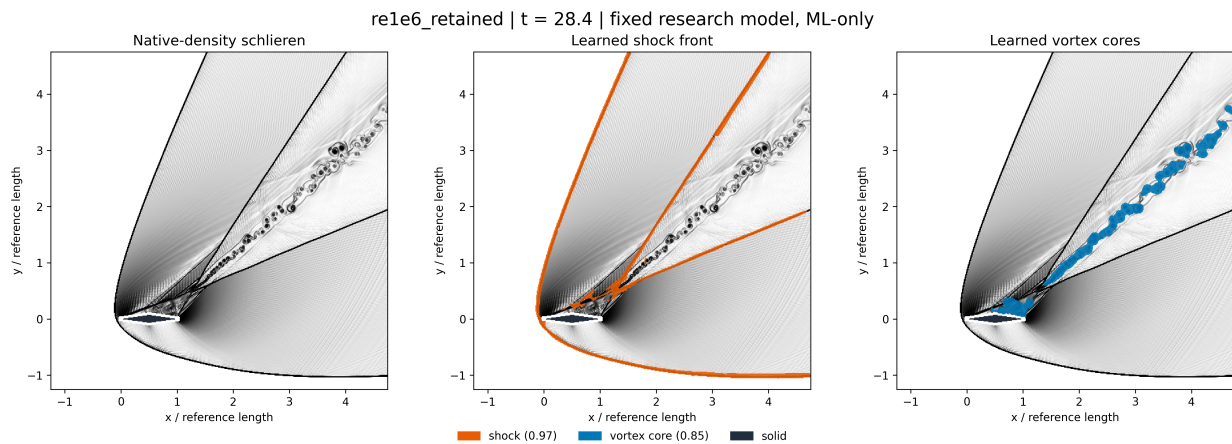
The notebook generates eight training cases, two validation cases and two held-out cases from different construction seeds. Entire fields define groups. Subsampling training pixels is allowed; randomly splitting adjacent pixels from one field between training and testing is not a test of case generalization. Research trajectories, grids and solver families need analogous grouping and duplicate-input checks.

Fit the feature scaler on training pixels only. The 32-by-24 tanh MLP uses a fixed optimizer budget; it is compared with compression and swirling-strength thresholds. Select each method's two operating thresholds on validation mean case-wise Dice, then freeze them. No test-dependent quantiles, loss-weight changes or mask cleanup are permitted.

Report convergence warnings rather than hiding them. A low training loss is not physical fidelity. A physical baseline can win; that is a valid result. The notebook's two test cases provide a classroom comparison, not a statistical generalization claim across flow families.

Assignment: remove rotational inputs, rebuild the model with a fresh predeclared split, and report both task metrics and changes in fragmentation. Never reuse the already-inspected classroom test as a new blind benchmark.

Read the colored outputs critically



Real airfoil field from Roohi's ShockVortexML study: native-density schlieren, learned shock mask and learned vortex mask. The fresh forward pass uses frozen thresholds 0.97 and 0.85. Geometry-crossing display-gradient stencils are excluded. These ML-only masks are not human labels or a hybrid. Thin missed branches and merged cores remain visible.

Dice = $2 * |\text{prediction} \cap \text{reference}| / (|\text{prediction}| + |\text{reference}|)$. Empty-versus-empty is assigned one in this lab; state the convention. A front shifted by one pixel can have poor Dice while remaining close physically. Conversely, a thick predicted envelope can gain overlap without locating the centerline accurately.

The gallery retains first/middle/last eligible times for both airfoil and cylinder, excluding all human-review frames. These are previously inspected development-test trajectories; do not infer independent accuracy from appearance or mask counts. The notebook retains the synthetic Dice exercise separately. All six full-size comparisons and checkpoint provenance are in [results/week11_research/](#).

Reference provenance and transfer gates

The manuscript uses physics-derived proposals as weak reference masks, not an independent exact oracle. If a hybrid uses those same proposals as support, its score measures reference-conditioned agreement. Learned-only and hybrid results must therefore remain separate. Development-inspected test fields are not untouched external tests.

Before transferring to research CFD, identify the source solver and case, native coordinates, solid mask, derivative stencil, nondimensional scales, reference construction, training-family membership and checkpoint. Preserve full-field arrays as well as visualizations. Walls and missing values must not become artificial vortices through interpolation or differentiation across the solid.

Add new controls for grid changes, noise and solver changes. Recompute derivatives from perturbed primitive velocities; perturbing only the already-derived diagnostic channels tests a different problem. A shock-overlap region must not automatically veto a real core. Thresholds expressed in raster pixels change physical size with resolution.

The research repository is <https://github.com/Ehsan-Roohi/ShockVortexML>. This lecture paraphrases the author-supplied manuscript and teaches its audit principles without copying a training pipeline or redistributing unpublished CFD archives.

Exit assessment and reproducibility

Submit one page with: the split IDs, frozen thresholds, both methods' case-wise Dice, at least one failure, and a labeled comparison figure. Explain why vorticity does not imply a vortex, why independent sigmoid labels are appropriate, and why physical-label agreement is not independent physical validation.

Propose a front-location metric and a component-matching rule before looking at a new research case. State how the rules handle geometry, ambiguity, missing structures and empty references. Describe the stronger evidence needed for cross-solver claims and for temporal vortex tracking.

Run notebooks/week11/W11_Shock_Vortex_Identification.ipynb from a complete FlowMLLab checkout or use its Colab launcher. It runs on CPU and keeps experiment outputs in memory. The authoring builder can regenerate the retained teaching notebook and figure; a student's Run All does not overwrite results/.

Source attribution: Ehsan Roohi, Physics-audited joint neural segmentation of shocks and vortex cores: cross-solver transfer and controlled airfoil--cylinder studies, author-supplied 2026 manuscript; ShockVortexML repository above. Original FlowMLLab lecture wording, synthetic fields and code were AI-assisted and checked. These additions are not claims of original research-data generation or full-paper reproduction.